A J-INVARIANT-CORRELATED CUBIC FINER-SPLIT IN POLYNOMIAL CONTINUED-FRACTION ASYMPTOTICS

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Abstract. For a polynomial continued fraction generated by an integer polynomial b of degree d, the convergent-error sequence has a WKB asymptotic with leading exponent $A_{\mathrm{PCF}}(b)$, whose sharp PCF-2 Conjecture B4 predicts $A_{\mathrm{PCF}} = 2d$. We report a degree-3 finer split: R1.1 found $\rho(\log(|j(E_b)|+1), A_{\mathrm{fit}} - 6) = -0.691$, and R1.3 reproduced it at deep precision with $\rho = -0.5683$, Bonferroni $p = 5.0 \times 10^{-5}$, on 50 cubic families. The four equianharmonic CM families in the $j = 0$ cell, with Weierstrass pattern $(1, -3, 3, *)$, hit $A = 6$ to $\leq 10^{-3}$. The sharp statement does not extend to degree 4: R13-C shows a real shallow-N $j = 0$ shift, but R13-D deep-WKB gives fam32 $\delta = -4.55 \times 10^{-4}$ and matched non-$j = 0$ fam01 $\delta = -4.60 \times 10^{-4}$, a 0.24 pooled-stderr difference. We therefore frame the phenomenon as cubic-modular, not as a cross-degree j-law; an Eisenstein/CM explanation is deferred to companion theory work. Companion records: PCF-1 v1.3 DOI 10.5281/zenodo.19937196; PCF-2 v1.2 DOI 10.5281/zenodo.19951458; Channel Theory v1.2 DOI 10.5281/zenodo.19951331; SIARC umbrella DOI 10.5281/zenodo.19885550.

1. Setup and verdict

Let $b(n) = a_d n^d + \cdots + a_0 \in \mathbb{Z}[n]$ be primitive and irreducible. The polynomial continued fraction

$$L(b) = b(0) + \cfrac{1}{b(1) + \frac{1}{b(2)+\cdots}}$$

has convergents $L_N = P_N/Q_N$ obtained from the usual second-order recurrence. The empirical PCF programme fits

$$\log|L_N - L| = -A_{\mathrm{PCF}}(b) N \log N + \alpha N + \beta \log N + \gamma + o(1),$$

and PCF-2's sharp B4 form predicts $A_{\mathrm{PCF}}(b) = 2d$ [3]. This note studies only the residual finer split $\delta_d(b) = A_{\mathrm{fit}}(b) - 2d$.

The verdict is deliberately narrow. At degree 3, $E_b : y^2 = b(x)$ is itself a nonsingular elliptic curve with its standard j-invariant; the observed signal is therefore a cubic-modular signal. We do not conjecture cross-degree universality in the sharp/deep-WKB form. Degree 4 still has a genus-one Jacobian and a j-invariant, but the R1.3 deep test rejects the analogous sharp rule. Claim labels such as R11-3 and R13-D-4 refer to AEAL entries in the bridge data citations [4, 5].

2. The cubic finer split

Theorem 1 (Empirical degree-3 verification)**.** *On the 50-family lex-window cubic catalogue, the j-invariant of E_b is a statistically significant predictor of $\delta_3 = A_{\mathrm{fit}} - 6$, while the $j = 0$ CM cell sits at the sharp value $A = 6$ to $\leq 10^{-3}$.*

(1) *R1.1 found $\rho(\log(|j|+1), \delta_3) = -0.6906$, raw $p = 2.86 \times 10^{-8}$, Bonferroni $p = 4.0 \times 10^{-7}$ with $K = 14$ effective tests; the signal also passed the stricter $K = 20$ accounting (*R11-3*).*
(2) *R13-A reproduced the signal at dps$= 2000$, $N \in [50, 250]$, $N_{\mathrm{ref}} = 400$: $\rho(\log(|j|+1), \delta_3^{\mathrm{free}}) = -0.5683$, raw $p = 1.67 \times 10^{-5}$, Bonferroni $p = 5.01 \times 10^{-5}$ (*R13-A-3*).*

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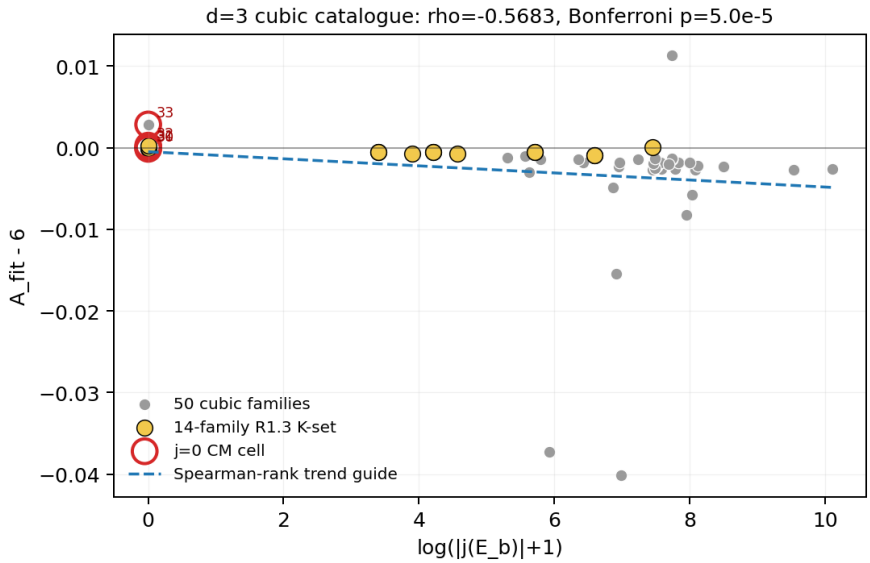

FIGURE 1. Cubic scatter of $\log(|j(E_b)|+1)$ against $A_{\text{fit}}-6$ from R13-A. Gray marks all 50 cubic families, gold marks the 14-family R1.3 diagnostic K-set generated by `make_figure.py`, and red open circles mark the four $j = 0$ equianharmonic CM families. The dashed line is a Spearman-rank trend guide for the negative association. This plot is $d = 3$ only; a $d = 4$ analog is shown in a separate companion plot to be deferred to a longer paper.

(3) *The four $j = 0$ families, equivalently the equianharmonic CM subcell with Weierstrass coefficient pattern $(1, -3, 3, *)$, are the closed-form candidates for $A = 6$; R1.1 precision escalation placed the cell at $\leq 10^{-3}$ and R1.3 retains the same cell-level interpretation (*R11-1*,* R13-A-3*).*

The figure is descriptive rather than a proof: the substantive thesis is not that the symbol j alone controls every PCF asymptotic, but that at $d = 3$ the PCF recurrence appears to see the modular structure of the elliptic curve E_b. The statistical statement is the Bonferroni-protected rank correlation; the cell statement is the special CM locus $j = 0$.

3. Precision-stratified non-extension to degree 4

The degree-4 analogue was tested explicitly in R1.3 and is not supported in sharp/deep-WKB form. On the original Q1 quartic catalogue, R13-B found no rank signal: $\rho(\log(|j| + 1), \delta_4^{\text{free}}) = +0.0732$, Bonferroni $p = 1$ (R13-B-6). The catalogue contains only one $j = 0$ quartic, so R13-C then enumerated an extended $j = 0$ cell. That shallow-N calculation found a real shift toward zero: after outlier removal, the $j = 0$ median was -2.493×10^{-3}, versus -3.727×10^{-3} for the Q1 non-$j = 0$ cluster, with Mann–Whitney $p = 4.8 \times 10^{-4}$ unstratified and $p = 1.1 \times 10^{-6}$ after stratifying on coefficient sum (R13-C-12, R13-E-25). This paragraph is part of the claim: the shallow effect is not hidden.

The referee-defense point is that R13-D then pushed the decisive comparator to deep WKB precision. At dps= 5000, $N_{\text{ref}} = 1000$, $N \in [200, 800]$, the unique Q1 $j = 0$ quartic family 32 gave

$$\delta_4^{\text{deep}}(\text{fam32}) = -4.5502 \times 10^{-4} \pm 1.45 \times 10^{-5},$$

while a matched non-$j = 0$ baseline, family 1, gave

$$\delta_4^{\text{deep}}(\text{fam01}) = -4.5993 \times 10^{-4} \pm 1.47 \times 10^{-5}.$$

The difference 4.91×10^{-6} is only 0.24 pooled standard errors (R13-D-1, R13-D-3, R13-D-4, R13-E-24). Thus the R13-C shift is a precision-stratified shallow-N diagnostic, not a deep leading-exponent rule. We explicitly disclaim a cross-degree statement of the form $j = 0 \Rightarrow A_{\mathrm{PCF}} = 2d$ or $\rho(\log |j|, \delta_d) < 0$ for all $d \geq 3$.

4. Cubic-modular interpretation

For cubics, the affine model $E_b : y^2 = b(x)$ is the elliptic curve whose j-invariant is measured. The cell $j(E_b) = 0$ is the equianharmonic CM point with endomorphism ring $\mathbb{Z}[\zeta_3]$. A natural interpretation is that the leading Birkhoff–Trjitzinsky/WKB exponent for the cubic PCF recurrence is sensitive to the Eisenstein modular structure of E_b, pinning the CM cell to the sharp value and inducing a monotone rank drift away from it.

This is still empirical. The degree-4 failure shows that a genus-one Jacobian and a numerical j-invariant are not sufficient, by themselves, to force the leading exponent. A proof would need to connect the recurrence's Newton polygon or channel-theory normal form to the Eisenstein CM specialization. The companion PCF-1, PCF-2, Channel Theory, and SIARC records give the surrounding framework [2, 3, 1, 6].

Open problem 1. Derive the degree-3 $j = 0$ value $A_{\mathrm{PCF}} = 6$ directly from the Birkhoff–Trjitzinsky expansion or an Eisenstein/CM normal form for E_b.

Open problem 2. Explain the R13-C shallow-N quartic shift as a sub-leading WKB term, and prove why it vanishes from the deep leading exponent.

Reproducibility and AEAL note

The numerical source files are the R1.1 and R1.3 bridge directories [4, 5]. R1.3 scripts include `r1_3_residualization.py`, `r1_3_extended_enumeration.py`, `r1_3_family32_deep.py`, and `r1_3_phase_E_final.py`; the concatenated R1.3 audit log contains 26 claims.[1] The DOI records cited here were checked through `doi.org` before packaging.

[1] Empirical claims in this note are documented as AEAL (Asynchronous Empirical Audit Log) claims with full statistical evidence at the bridge URLs in the bibliography. Each claim_id can be queried for its full provenance.

References

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[4] papanokechi. PCF2-SESSION-R1.1: cubic finer-split discovery source files and AEAL claims. `https://github.com/papanokechi/siarc-relay-bridge/tree/main/sessions/2026-05-01/PCF2-SESSION-R1.1/`, 2026. Bridge data citation.

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[6] papanokechi. SIARC umbrella record. Zenodo, 2026. DOI: 10.5281/zenodo.19885550.